

BRIEF REPORT

Changes in Coronary Physiology by μ FR Measurement After DREAMS-3G-Scaffold Implantation



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Bioresorbable stent scaffolds provide a potential alternative approach that avoids the long-term presence of a rigid cage that may prevent restoration of coronary artery function and physiology. The DREAMS 3G resorbable magnesium scaffold safety outcomes have been previously demonstrated in the BIOMAG (BIOTRONIK - Safety and Clinical Performance of the Sirolimus-Eluting Resorbable Coronary Magnesium Scaffold System [DREAMS 3G] in the Treatment of Subjects with de Novo Lesions in Native Coronary Arteries)-1 trial.¹ The third-generation DREAMS 3G scaffold system is composed of resorbable magnesium alloy that resorbs within 12 months and most notably has increased radial strength and thinner struts.² Given this dynamic process, the affected vessel undergoes

significant changes in vessel support and flexibility over a 12-month period of resorption. While previous studies have focused on improvements in in-scaffold late lumen loss (LLL) between the DREAMS 3G and previous generation scaffold system, this dynamic process may be better reflected by nonanatomical outcomes.²

We performed a post hoc analysis of the BIOMAG-1 study to investigate the temporal changes in non-invasive Murray law-based quantitative flow ratio— μ FR (Pulse Medical Technology Inc) measurements after DREAMS 3G implantation and compare them to in-scaffold LLL over the same time frame. Local ethics committees at participants' sites approved the study protocol.

In the BIOMAG-1 trial, a total of 116 patients were enrolled. Full inclusion and exclusion criteria can be found in the original BIOMAG-1 trial.¹ μ FR is a non-invasive Murray law-based quantitative flow ratio using one view in this study. Patients underwent μ FR assessment before and immediately after DREAMS 3G implantation during index procedure. μ FR was also measured at follow-up 6- and 12-month angiographic reassessment. A hemodynamically significant μ FR of clinical importance is ≤ 0.8 . An independent core laboratory (MedStar Cardiovascular Research Network) performed the analysis. In-scaffold LLL was

What is the clinical question being addressed?

Is angiography-derived physiology changes feasible and clinically meaningful?

What is the main finding?

μ FR measurements significantly improved to clinically normal levels after DREAMS 3G scaffold implantation, and these levels of μ FR improvement are sustained to 12-month follow-up.

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

ABBREVIATIONS AND ACRONYMS

LLL = late lumen loss

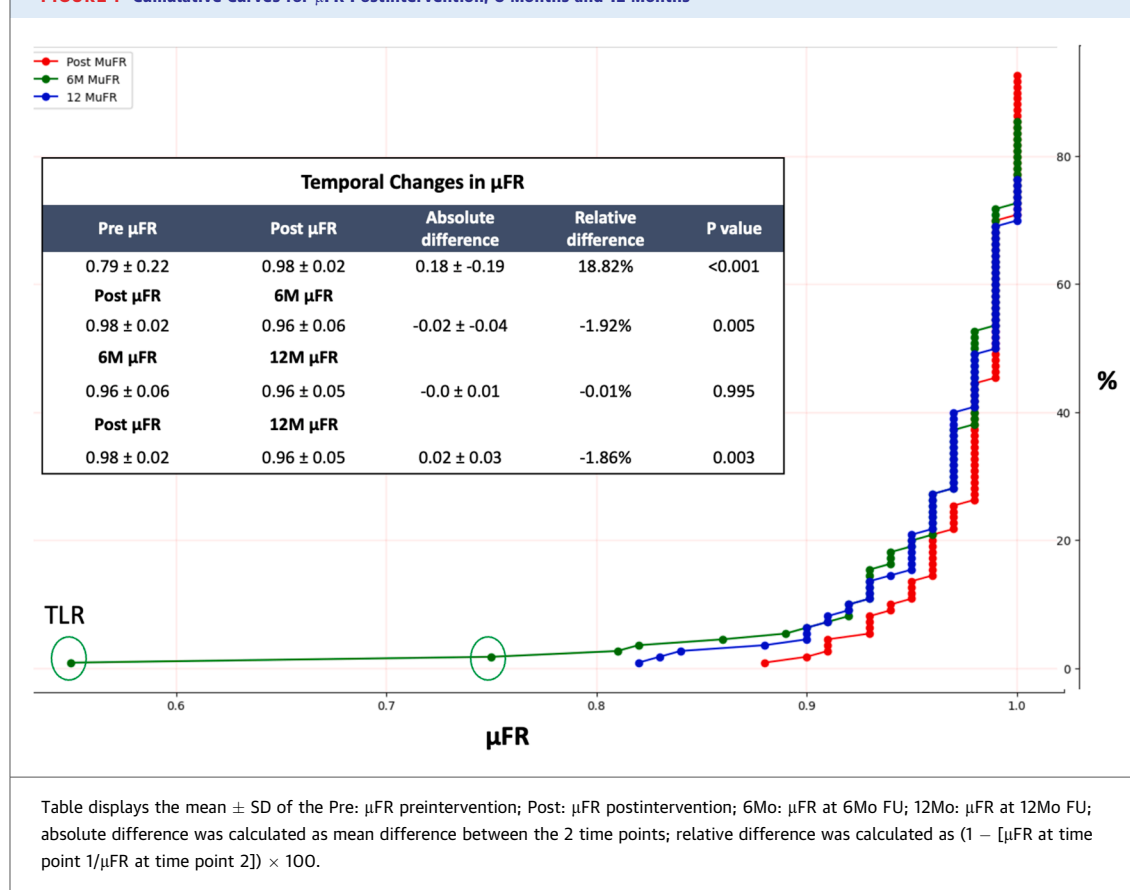
measured by quantitative coronary angiography and was assessed at 6 months and 12 months.

After normality testing was applied, the μ FR measurements between different time points were compared using a 2-tailed *t*-test. The level of significance (alpha) was set at 0.05 for all statistical tests. Absolute difference was calculated as the mean difference between μ FR measurements at various time intervals. Relative difference was also obtained by calculating $(1 - [\mu\text{FR at time point 1}/\mu\text{FR at time point 2}]) \times 100$, representing a percent change between the time intervals. Coefficient correlations between μ FR and LLL were calculated. All statistical analyses were performed using Jupyter Lab version 3.4.4 based on Python version 3.9.13 and SAS (IBM) 9.4.

The full baseline demographics of the patient population and lesion characteristics can be found in

the original manuscript.¹ After implantation of DREAMS 3G, μ FR increased from a clinically abnormal mean measurement of 0.79 ± 0.22 to a clinically normal mean measurement of 0.98 ± 0.02 ($P < 0.001$). Compared to baseline μ FR, there was a decrease in μ FR at 6 months (0.98 ± 0.02 vs 0.96 ± 0.06 ; $P = 0.005$), but it remained above the clinically abnormal μ FR threshold of ≤ 0.8 . The μ FR measurements compared between 6 and 12 months showed no significant difference (0.96 ± 0.06 vs 0.96 ± 0.05 ; $P = 0.995$). The μ FR immediately post DREAMS 3G implantation compared to 12-month follow-up was significantly reduced (0.98 ± 0.02 vs 0.96 ± 0.05 ; $P = 0.003$) but remained well above the clinically abnormal μ FR threshold of ≤ 0.8 . Relative difference in μ FR immediately post DREAMS 3G implantation compared to 12 months showed a 1.86% decrease in μ FR, after an initial increase in μ FR of 18.82% immediately postimplantation. See [Figure 1](#)

FIGURE 1 Cumulative Curves for μ FR Postintervention, 6 Months and 12 Months



for more details regarding temporal changes in μ FR. Of the patients included in the analysis, all but 3 patients maintained a μ FR above 0.80 over 12-month follow-up (Figure 1). These patients had a target lesion revascularization procedure. μ FR measurement prior to bioresorbable scaffold did not correlate with LLL at either 6-month ($r = -0.040$) or 12-month follow-up ($r = 0.08$). Furthermore, μ FR measurement had a negative correlation when comparing both 6-month μ FR with 6-month LLL ($r = -0.48$), as well as 12-month μ FR with 12-month LLL (-0.53).

Previous anatomical assessment of this same cohort has shown that while there is lower scaffold LLL compared to the previous bioresorbable scaffold generation, there remains a scaffold LLL of 0.24 ± 0.36 mm (95% CI: 0.02-0.19) over a 12-month follow-up period.¹ However, LLL may not always predict clinical outcomes,³ and previous studies have shown strong correlation between initial improvement in post-percutaneous coronary intervention fractional flow reserve and patient quality of life improvements and angina.⁴ In our study that included stable patients with strict inclusion and exclusion criteria and had excluded large bifurcations, heavy calcifications, and acute myocardial infarction, μ FR measurement was significantly improved from preimplantation to immediately postimplantation, and although there was a statistically significant decrease in μ FR at 12-month follow-up, all μ FR measurements remained well above the clinically accepted abnormal threshold of μ FR ≤ 0.80 . Furthermore, there was a moderate negative correlation between the μ FR and late luminal loss at 6 and 12 months. In addition to correlation with symptom relief, coronary physiology may better reflect overall vessel health compared to anatomical outcomes.

In conclusion, our findings suggest that μ FR is significantly improved to clinically normal levels after DREAMS 3G scaffold implantation and that these levels of μ FR improvement are sustained to 12-month follow-up.

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